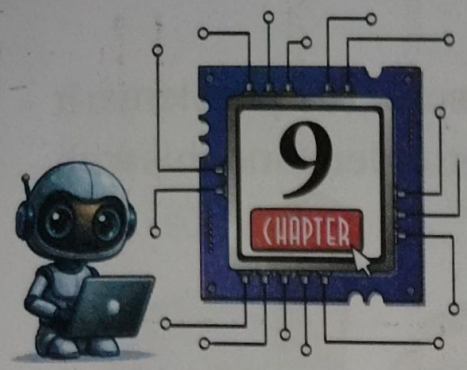


Introduction to AI and Robotics



Let's Recall

(i) If Rohan has 3 toffees and Mohan has 5 toffees, how many toffees do they have in total?



(ii) Count the number of fruits in the basket and write their names along with how many of each are there.





In this Chapter



- What is Intelligence?
- Concept of AI
- Concept of Robot



Intelligence is the ability to learn, understand and solve the problems. It helps you to think, reason and make decisions. Let us understand this with the help of an example :



Akash is standing on a traffic signal.
The signal shows the red light.

Should he cross the road or not?

Yes ☐ No ☐

This ability to think about a situation and make correct decision is called intelligence.

ARTIFICIAL INTELLIGENCE

We have the ability to think and make decisions. Similary, if machines can learn and make decisions on it's own then it is called Artificial Intelligence.

Let us look at some examples of artificial intelligence :

Smart phones have voice assistants like siri and google assistant. You can talk to them. They will answer your questions and help you with tasks.



Voice Assistant



Smart Watch

Now-a-days, we have smart watches also. They perform various tasks like tracking and counting your steps and many more.

Smart Camera is an intelligent device. It can detect any movement and can rotate automatically.



Smart Camera

Smart doors open and close automatically, when someone comes near the door. They senses movement to do so.



Smart Doors



Smart Speaker

You can communicate with a smart speaker. It listens to the command when you speak and responds. You can hear songs on it.

Smart cars run automatically without any driver. They are safe and useful.



Smart Car

WHAT IS A ROBOT?

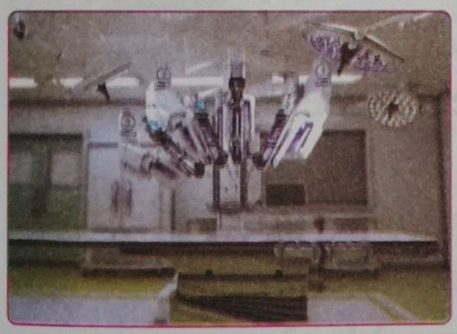
A robot is a machine that can do tasks on it's own. It can move, sense and perform actions.

The brain is more powerful than any intelligent machine. Intelligent machines like robot are made by man. Let us look at some types of robots.

Smart robot can play music, dance and tell stories.

Roomba is a robotic vacuum cleaner. It cleans the floor of a house automatically.





Da Vinci Surgical system robot is used in hospitals to help doctors in operations.



This is ASIMO robot. It can walk, talk and interact with people.

These are flying robots used for taking photos, delivering packages, etc. They are called drones.





This is a robotic dog called AIBO. It is a smart toy. It can play and talk with you.

Let's Get Better

How can robots and AI help us in our daily lives?



I. Tick (✓) the correct option :

1. Which of the following has natural intelligence?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

2. Which of the following has artificial intelligence?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

3. Which of the following is an intelligent machine?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

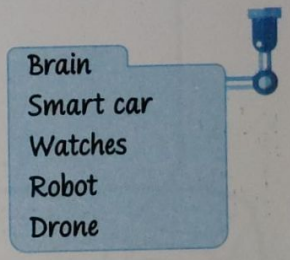
4. Which of the following is an intelligent device?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

II. Match the following :

1. 	•	• (a) Flying Robot
2. 	•	• (b) Smart Camera
3. 	•	• (c) Toy Robot
4. 	•	• (d) Smart Speaker

III. Fill in the blanks with the help of the box :



1. On smart ..... you can dial or receive the calls.
2. We use our ..... and intelligence to find the answer.
3. ..... camera is called flying robot.
4. ..... runs without driver.
5. ..... needs orders or commands from humans to do any tasks.